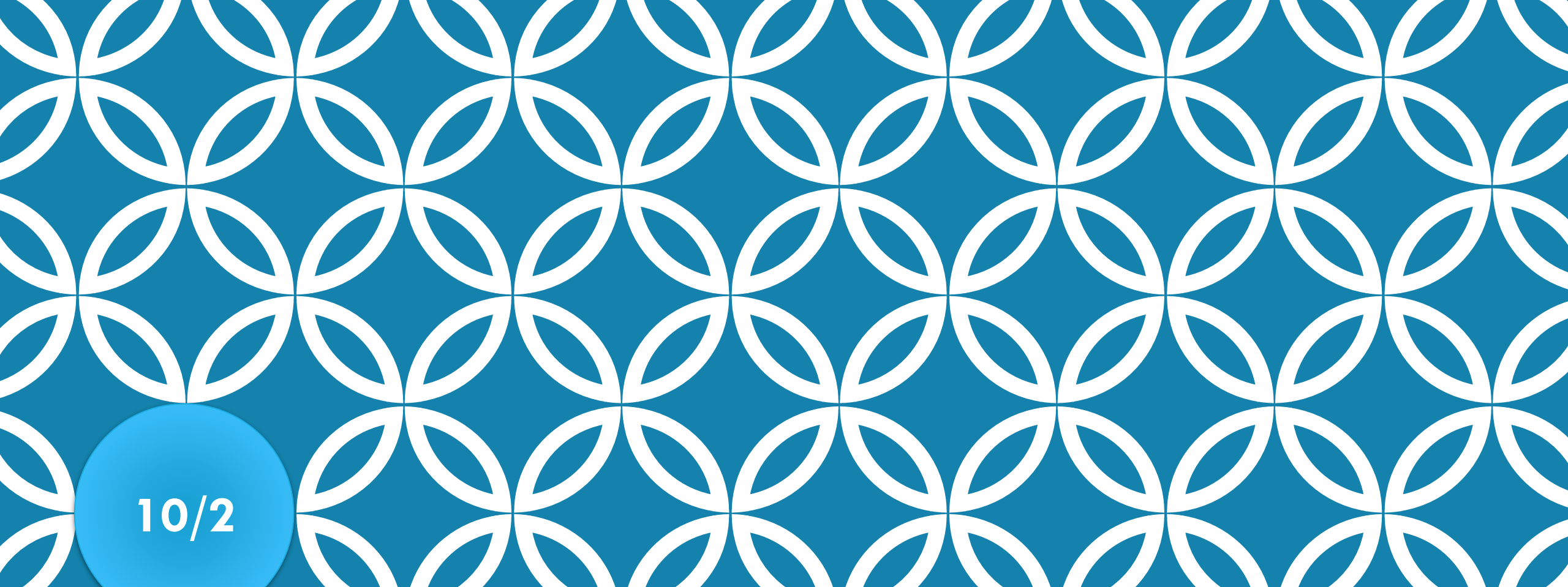




10/2

1st-order logic: Semantics

Interpretation
Assignment
Evaluation

Logic in Computer Science Lab
Gyöngyi Bujdosó
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Based on prof. Tamás Kádek's lectures

Interpretation

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language. The pair $\langle U, \varrho \rangle$ is the *interpretation* of $L^{(1)}$ if

- U is a nonempty set of objects; and
- ϱ is a mapping such that $Dom(\varrho) = Con$ and
 - if $a \in \mathcal{F}(0)$, akkor $\varrho(a) \in U$;
 - if $f \in \mathcal{F}(n)$ ($n > 0$), then

$$\varrho(f) \in U^{U^n} \text{ and so } \varrho(f) : U^n \rightarrow U;$$

- if $p \in \mathcal{P}(0)$, akkor $\varrho(p) \in \{0, 1\}$;
- if $P \in \mathcal{P}(n)$ ($n > 0$), then

$$\varrho(P) \in \{0, 1\}^{U^n} \text{ and so } \varrho(P) : U^n \rightarrow \{0, 1\}.$$

Assignment

Variable assignment

Let $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language and $\langle U, \varrho \rangle$ be an interpretation of $\mathcal{L}^{(1)}$. The v mapping is an *assignment* relying on the interpretation $\langle U, \varrho \rangle$ if:

$$v : Var \rightarrow U$$

Modified assignment

Let v be an assignment relying on the interpretation $\langle U, \varrho \rangle$, $x \in Var$ be a variable and $u \in U$ be an object. If it is true for all $y \in Var$ that

$$v[x : u](y) = \begin{cases} u, & \text{if } y = x; \\ v(y), & \text{otherwise.} \end{cases}$$

then $v[x : u]$ is a *modified assignment*. The $v[x : u]$ function is an *x-variant* of v .

Evaluation of a term

Let $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language, $\langle U, \varrho \rangle$ be one of its interpretations and v be an assignment relying on $\langle U, \varrho \rangle$.

- If $a \in \mathcal{F}(0)$, then $|a|_v^{\langle U, \varrho \rangle} = \varrho(a)$.
- If $x \in Var$, then $|x|_v^{\langle U, \varrho \rangle} = v(x)$.
- If $f \in \mathcal{F}(n)$ for some $n > 0$, then

$$|f(t_1, \dots, t_n)|_v^{\langle U, \varrho \rangle} = \varrho(f)(|t_1|_v^{\langle U, \varrho \rangle}, \dots, |t_n|_v^{\langle U, \varrho \rangle})$$

where $t_i \in Term$ for all $i \in \{1, 2, \dots, n\}$.

Evaluation of an Atomic Formula

- If $p \in \mathcal{P}(0)$, then $|p|_v^{\langle U, e \rangle} = \varrho(p)$
- If $t_1, t_2 \in Term$, then

$$|(t_1 = t_2)|_v^{\langle U, e \rangle} = \begin{cases} 1, & \text{if } |t_1|_v^{\langle U, e \rangle} = |t_2|_v^{\langle U, e \rangle} \\ 0, & \text{otherwise.} \end{cases}$$

- If $P \in \mathcal{P}(n)$ for some $n > 0$, then

$$|P(t_1, \dots, t_n)|_v^{\langle U, e \rangle} = \varrho(P)(|t_1|_v^{\langle U, e \rangle}, \dots, |t_n|_v^{\langle U, e \rangle})$$

where $t_i \in Term$ for all $i \in \{1, 2, \dots, n\}$.

Evaluation of a Compound Formula 1

Evaluation rules inherited from zero-order logic

- If $A \in Form$, then $| \neg A |_v^{\langle U, e \rangle} = 1 - | A |_v^{\langle U, e \rangle}$.
- If $A, B \in Form$, then

$$| (A \supset B) |_v^{\langle U, e \rangle} = \begin{cases} 0 & \text{if } | A |_v^{\langle U, e \rangle} = 1, \text{ and } | B |_v^{\langle U, e \rangle} = 0; \\ 1, & \text{otherwise.} \end{cases}$$

$$| (A \wedge B) |_v^{\langle U, e \rangle} = \begin{cases} 1 & \text{if } | A |_v^{\langle U, e \rangle} = 1, \text{ and } | B |_v^{\langle U, e \rangle} = 1; \\ 0, & \text{otherwise.} \end{cases}$$

$$| (A \vee B) |_v^{\langle U, e \rangle} = \begin{cases} 0 & \text{if } | A |_v^{\langle U, e \rangle} = 0, \text{ and } | B |_v^{\langle U, e \rangle} = 0; \\ 1, & \text{otherwise.} \end{cases}$$

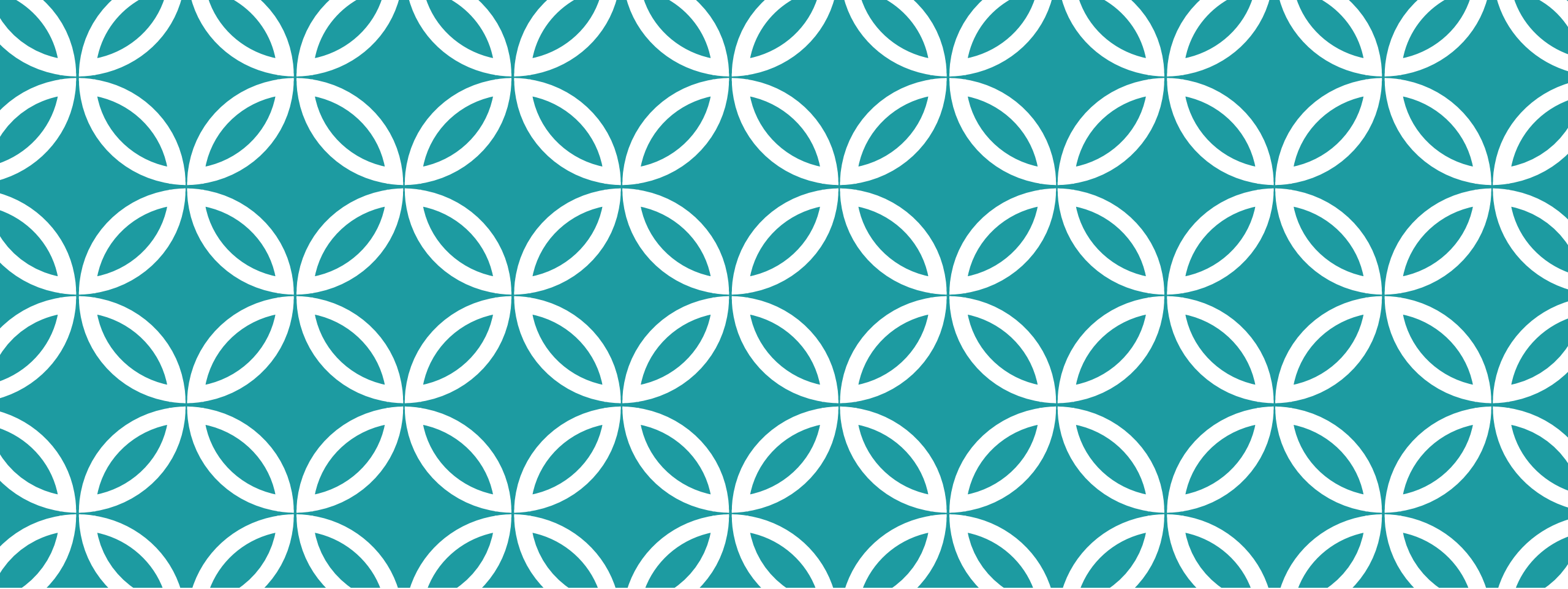
$$| (A \equiv B) |_v^{\langle U, e \rangle} = \begin{cases} 1 & \text{if } | A |_v^{\langle U, e \rangle} = | B |_v^{\langle U, e \rangle} \\ 0, & \text{otherwise.} \end{cases}$$

Evaluation of a Compound Formula 2

- If $A \in \text{Form}$, $x \in \text{Var}$, then

$$|\forall x A|_v^{\langle U, \varrho \rangle} = \begin{cases} 0, & \text{if there exists an } u \in U, \text{ such that } |A|_{v[x:u]}^{\langle U, \varrho \rangle} = 0; \\ 1, & \text{otherwise.} \end{cases}$$

$$|\exists x A|_v^{\langle U, \varrho \rangle} = \begin{cases} 1, & \text{if there exists an } u \in U, \text{ such that } |A|_{v[x:u]}^{\langle U, \varrho \rangle} = 1; \\ 0, & \text{otherwise.} \end{cases}$$



Exercises

3. What is the semantic values of the following terms, if

$U = \mathbb{Z}$ (set of the whole numbers), $\varrho(c) = 4$, $\varrho(d) = 3$, $\varrho(f)(n) = \lfloor \frac{n}{2} \rfloor$ (lower integer part), $\varrho(g)(n, m) = n \times m$, and $v(x) = 10$.

(a) $f(g(f(f(x))), d)$

(b) $g(g(d, f(d)), d)$

(c) $g(f(c), x)$

(d) $f(g(c, f(d)))$

(e) $g(g(c, c), c)$

(f) $g(c, f(f(x)))$

3. What is the semantic values of the following terms, if

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(a) $f(g(f(f(x))), d)$

(a) _____ 3 _____

(b) $g(g(d, f(d)), d)$

(b) _____ 9 _____

(c) $g(f(c), x)$

(c) _____ 20 _____

(d) $f(g(c, f(d)))$

(d) _____ 2 _____

(e) $g(g(c, c), c)$

(e) _____ 64 _____

(f) $g(c, f(f(x)))$

(f) _____ 8 _____

2.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = \langle LC, \{x_1, x_2, \dots\}, \{P, Q, c\}, Term, Form \rangle$$

$$P \in \mathcal{P}(1), \quad Q \in \mathcal{P}(2), \quad c \in \mathcal{F}(0).$$

Let $\langle U, \varrho \rangle$ az $L^{(1)}$ be an interpretation of the language, so that:

- $U = \{u_1, u_2, u_3\}$
- $\varrho(P) = P' \quad P'(x) = \begin{cases} 1 & \text{if } x = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(Q) = Q' \quad Q'(x, y) = \begin{cases} 1 & \text{if } x = u_1 \text{ és } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation, so that $v: Var \rightarrow U$ and

$$v(x_1) = u_3, \quad v(x_2) = u_1$$

Which of the formulas below are true?

$$|(Q(c, c) \equiv P(x_2))|_v^{\langle U, \varrho \rangle}$$

$$|\neg P(x_1)|_v^{\langle U, \varrho \rangle}$$

$$|(Q(c, c) \equiv P(x_1))|_v^{\langle U, \varrho \rangle}$$

$$|(\neg P(x_1) \wedge P(c))|_v^{\langle U, \varrho \rangle}$$

$$|(Q(x_1, x_2) \supset P(x_1))|_v^{\langle U, \varrho \rangle}$$

$$|((x_1 = x_2) \supset (x_2 = c))|_v^{\langle U, \varrho \rangle}$$

$$|(P(x_2) \supset Q(x_1, x_2))|_v^{\langle U, \varrho \rangle}$$

2.) Let $L^{(1)}$ be a first-order language defined as follows:

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$$|(Q(x_1, x_2) \supset P(x_1))|_v^{\langle U, \varrho \rangle}$$



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$$|(P(x_2) \supset Q(x_1, x_2))|_v^{\langle U, \varrho \rangle}$$

Which of the following formulae are true, if $U=N$ (set of the natural numbers):

- $q(c) = 2$
- $q(d) = 1$
- $q(f)(n) = 2 \times n$
- $q(g)(n,m) = n+m$
- $q(P)(n) = n$ is odd
- $q(Q)(n,m) = n < m$
- $v(x) = 2$
- $v(y) = 3$

1. $x = x$
2. $Q(d, f(g(f(c), d)))$
3. $Q(c, c) \wedge Q(f(c), g(c, f(c))) \equiv Q(c, d)$
4. $P(d) \supset c = d$
5. $P(f(f(g(c, c))))$
6. $g(g(c, c), g(d, f(c))) = g(d, d)$
7. $\neg Q(d, d)$
8. $\forall x \neg P(f(x))$
9. $f(c) = f(x) \equiv \neg c = d$
10. $Q(c, f(d)) \wedge Q(f(c), f(g(c, g(c, c))))$

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7. $\neg Q(d, d)$
8. $\forall x \neg P(f(x))$
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10. $Q(c, f(d)) \wedge Q(f(c), f(g(c, g(c, c))))$

4. Which of the following formulae are true, if

$U = \mathbb{N}$ (set of the natural numbers), $\varrho(c) = 2$, $\varrho(d) = 1$, $\varrho(f)(n) = 2 \times n$, $\varrho(g)(n, m) = n + m$, $\varrho(P)(n) = n$ is odd, $\varrho(Q)(n, m) = n < m$, $v(x) = 2$ and $v(y) = 3$.

- ☐ $x = x$ ☐ $Q(d, f(g(f(c), d)))$ ☐ $x = y$ ☐ $Q(c, c) \wedge Q(f(c), g(c, f(c))) \equiv Q(c, d)$ ☐ $c = d$
☐ $P(d) \supset c = d$ ☐ $P(f(f(g(c, c))))$ ☐ $g(g(c, c), g(d, f(c))) = g(d, d)$ ☐ $Q(d, c)$ ☐ $\neg Q(d, d)$
☐ $c = c \equiv f(c) = g(c, f(c)) \vee P(c)$ ☐ $\neg P(d)$ ☐ $\exists x P(g(x, x))$ ☐ $\neg g(c, g(c, d)) = y$ ☐ $d = x \supset d = g(c, c)$
☐ $\exists x (d = x \supset d = g(c, c))$ ☐ $\forall x (d = x \supset d = g(c, c))$ ☐ $f(c) = f(x) \equiv \neg c = d$
☐ $\forall x \neg P(f(x))$ ☐ $\neg \exists x P(f(x))$ ☐ $Q(c, f(d)) \wedge Q(f(c), f(g(c, g(c, c))))$ ☐ $Q(c, f(c)) \vee P(c)$
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- ☒ $x = x$ ☒ $Q(d, f(g(f(c), d)))$ ☐ $x = y$ ☒ $Q(c, c) \wedge Q(f(c), g(c, f(c))) \equiv Q(c, d)$ ☐ $c = d$
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- $q(g)(n,m) = n-m$
- $q(P)(n) = n$ is positive
- $q(Q)(n,m) = n \neq m$
- $v(x) = 1$
- $v(y) = -1$

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